Assignment 01: Part-of-Speech (POS) Extraction from a News Article

Objective:
Apply Natural Language Processing techniques to identify and extract nouns and verbs from an English news article.

Tasks

1. Download any **English-language news article** from a reputable news source.
2. Save the article content as a **plain text (.txt) file**.
3. Load the text file into a Python program or Jupyter Notebook.
4. Perform **Part-of-Speech (POS) tagging** on the text.
5. Extract all **Nouns** and **Verbs** from the article.
6. Save the extracted words into a **CSV file** with appropriate columns (e.g., `Word`, `POS_Tag`).
7. Submit your solution as a Jupyter Notebook.

Submission Requirements

- **Notebook File Name Format:**
  `YourName_POS_Identifier.ipynb`

  Example:
  `PremBist_POS_01.ipynb`

- **Output File:**
  `YourName_POS_Identifier.csv`

Deliverables

- Jupyter Notebook (`.ipynb`)
- Original news article text file (`.txt`)
- CSV file containing the extracted nouns and verbs (`.csv`)

Evaluation Criteria

- Correct loading and processing of the text file
- Accurate POS tagging and extraction of nouns and verbs
- Proper CSV export
- Code readability, documentation, and organization
- Adherence to the required file naming convention

Assignment 02: Named Entity Recognition (NER) from a News Article

Objective:
Apply Named Entity Recognition (NER) techniques to identify and extract named entities from an English news article.

Tasks

1. Download any **English-language news article** from a reputable news source.
2. Save the article content as a **plain text (.txt) file**.
3. Load the text file into a Python program or Jupyter Notebook.
4. Perform **Named Entity Recognition (NER)** on the text using an NLP library such as **spaCy**, **NLTK**, or **Hugging Face Transformers**.
5. Extract all named entities and their corresponding entity types (e.g., `PERSON`, `ORG`, `GPE`, `DATE`, `MONEY`, etc.).
6. Save the extracted entities into a **CSV file** with appropriate columns (e.g., `Entity`, `Entity_Type`).
7. Provide a summary of the different entity types found in the article.
8. Submit your solution as a Jupyter Notebook.

Submission Requirements

- **Notebook File Name Format:**
  `YourName_NER_Identifier.ipynb`

  Example:
  `PremBist_NER_01.ipynb`

- **Output File:**
  `YourName_NER_Identifier.csv`

Deliverables

- Jupyter Notebook (`.ipynb`)
- Original news article text file (`.txt`)
- CSV file containing the extracted named entities (`.csv`)
- A brief summary of entity counts by category

Example CSV Format

Entity	Entity_Type
Donald Trump	PERSON
United States	GPE
Microsoft	ORG
July 20, 2026	DATE

Evaluation Criteria

- Correct loading and preprocessing of the text file
- Accurate implementation of Named Entity Recognition
- Proper extraction and labeling of entities
- Correct CSV export and formatting
- Quality of code, documentation, and analysis
- Adherence to the required file naming convention

Bonus (Optional):

- Visualize the recognized entities using spaCy's `displacy` or another visualization tool.
- Compare the performance of two different NER models and discuss the results.